

Medical Diagnosis AID

Predicting diabetes likelihood from clinical data

Let's Explore Together!

- Dataset and the challenge
- EDA and preprocessing
- Models and results
- Demo MVP

Dataset and the challenge

Dataset: PIMA Indians Diabetes Database

Task: Binary Classification

Target: Outcome

768 rows, 8 input features, 1 target





Exploratory Data Analysis and preprocessing

The dataset was inspected before training.

Some zero values were medically improbable, so they were replaced with NaN and then imputed with the median.

Not every zero was treated as missing: for example, pregnancies = 0 is valid.

Models and results

Model	Accuracy	Precision	Recall	F1-score*
Logistic Regression	0.701	0.587	0.500	0.540
Random Forest Cl.	0.779	0.727	0.593	0.653

Logistic Regression was used as a baseline model.

RandomForestClassifier was chosen as the final model because it performed better on the test set across all main metrics.

Demo MVP

MEDICAL DIAGNOSIS AID

Diabetes Risk Prediction

This tool estimates the likelihood of diabetes from basic clinical measurements. Enter the patient data below and submit the form to get a prediction.

Important: This application is for educational purposes only. If you do not know these values, use recent medical results or ask a healthcare professional instead of guessing.

Pregnancies Total number of previous pregnancies.	Glucose Plasma glucose concentration measured during a clinical test.
Blood Pressure Diastolic blood pressure in mmHg, usually measured during a medical visit.	Skin Thickness Triceps skin fold thickness in mm, typically measured by a health professional.
Insulin 2-hour serum insulin level from a laboratory test.	BMI Body Mass Index, usually computed from height and weight.
Diabetes Pedigree Function A dataset score related to family history and hereditary diabetes risk.	Age Patient age in years.

Predict

Flask web app with HTML form

Prediction result page

JSON API route

Docker support

Let's Take off Together!

Simple project, but fully end-to-end.

Tailor-made with love to be
Understandable,
Reproducible and
Defendable.